School of Technology, Engineering, Mathematics and Physics

IS222 Database Management Systems

Semester 1, 2021

Mid-Semester Test – Solutions

Learning Outcomes:

1. **Explain the fundamental concepts of various database management systems.**
2. **Construct a data model for a relational database management system.**

Total marks : **100 marks (25% of coursework)**
Date/Day : **Friday 21 May, 2021**
Duration : **6 hours (Start Time : 9:00 am, End Time : 3:00 pm) (Fiji Standard Time)**

This examination has two Parts:

Part A: Online Quiz (25 marks -7%)
Part B: Problem-based Questions (75 marks -18%)

MST Part A - Online Quiz
Instructions:

- This quiz is to be attempted online in Moodle via the MST Online Quiz link.
- There are 25 questions. All questions are compulsory.
- This test consists of multiple-choice, true-false, matching, and short answers.
- The total mark is 25 and weighs 7% towards the coursework.
- You are given 60 minutes to complete this online quiz.
- You can attempt this quiz only once. Please note that no extra attempts will be allowed for the quiz.
- You can attempt the quiz anytime between the start and 60 minutes prior to the end time of the MST.
- Please click on Submit All and finish button when you have answered all questions to have your quiz graded.

MST Part B – Problem-based questions
Instructions:

- There are 8 questions. All questions are compulsory.
- This test consists of a case study followed by a set of related problem-based questions.
- The total mark is 75 and weighs 18% towards the coursework.
- Type your answers to part B in the special Answer Booklet MS Word document provided under the *MST Test Information* section in the *Course Assessment* Block in Moodle. Save the Answer Booklet using your student id number. For example: SXXXXXXXX.docx where X's represent your id number.

- Submit your Answer Booklet document file in the **MST Part B dropbox** provided in Moodle by the MST end time. DO NOT convert your document to pdf or any other format.
- Note Turnitin is enabled and similarity index of above 20% will be considered as plagiarism.
- You can attempt this test only once. Please note that no extra attempts will be allowed for the test. Thus, ensure that you are submitting the correct file.
- You can only submit one file – your MS word Answer Booklet document.
- You must attempt the test questions and submit before the MST end time.

Note:

- Both parts A and B of the MST must be completed within the allocated duration of the MST.
- Ensure that you have reliable and consistent internet access to attempt the MST.

MST Part B – Problem-based Questions

Identifying anomalies in relations (8 marks)
Consider the following table:

Project Number	Project Name	Employee Number	Employee Name	Department	Hours Spent
1	Apollo	1001	Smith	Engineering	127
1	Apollo	1003	Jones	Accounts	45
2	Mercury	1002	Stewart	Marketing	70
3	Venus	1001	Smith	Engineering	21
3	Venus	1006	Brown	Engineering	124

1. Identify the potential problems that may arise if the data was kept in this relation. *(3 marks)*

Data redundancy (duplication) leading to insert, update and delete anomalies.

2. Design a set of relational schemas that would avoid the problems you identified. Ensure you clearly identify the primary key in each schema, and any foreign keys. *(5 marks)*

   Employee (Employee Number (PK), Employee Name, Department ID (FK))
   Project (Project Number (PK), Project Name)
   Department (Department ID (PK), Department Name)
   ProjectStaff (Project Number (PK) (FK), Employee Number (PK) (FK), Hours Spent)

Relational Algebra (15 marks)
Use the following tables to answer question 3.

Table name: STUDENT

STU_CODE	PROF_CODE
100278	
128569	2
512272	4
531235	2
531268	
553427	1

Table name: PROFESSOR

PROF_CODE	DEPT_CODE
1	2
2	6
3	6
4	4

3. Perform the following Relations Algebra on the above two tables and show the result:
   a) Natural Join *(5 marks)*

STU_CODE	PROF_CODE	DEPT_CODE
128569	2	6
512272	4	4
531235	2	6
553427	1	2

b) Left outer join *(5 marks)*

RightOuterJoin : Select Query

STU_CODE	PROF_CODE	DEPT_CODE
100278		
531268		
553427	1	2
128569	2	6
531235	2	6
512272	4	4

Record: 1 of 6

c) Right outer join *(5 marks)*

LeftOuterJoin : Select Query

STU_CODE	PROF_CODE	DEPT_CODE
553427	1	2
128569	2	6
531235	2	6
	3	6
512272	4	4

Record: 1 of 5

Entity-Relationship Modelling (22 marks)
Use the following case study description to answer questions 4 and 5.

Smiggle Corporation (SC) has a department store with average of 100 customers per day. To perform more efficiently, Smiggle Corporation has decided to develop a database system to run day-to-day transactions of the department store. Such system is also referred to as on-line transaction processing (OLTP) systems, where the database is updated continually throughout the day. The transactions should be captured at the point of sales, assuming one record for each purchased item. Each customer averages 5 items per transaction. Thus, in average 5000 records will be added to the department store database every day.

You are contracted to work on the Smiggle's Sales department store database project and take it through the stages of design and implementation. You are to analyze the requirements for, design, implement, and demonstrate a database system that could automate the functions of the department store. The result of your assignment should be a working implementation that provides access to information in a relational database.

Your database should maintain a current record of inventory that is updated to reflect the transactions performed. You must keep track of customers, including their personal information, and purchase history. For customers you should keep records about customer ID, name, address (street, city, zip code), marriage status, gender, and age. For products you should maintain information about product ID, product name, inventory amount, price, product category, product manufacturer. Currently SC handles the following product categories: food, electronics, sport equipment, and office supply. Those products are sold in corresponding departments of the store. Your database should maintain information about store employees, including employee ID, name, address, e-mail, job position (salesperson, or manager), department assigned, and salary.

Sale transactions reflect records of product purchased, including order number, date of purchase, salesperson ID, product information (sale price, quantity), customer information and department ID where the transaction is completed. Information about each department includes department ID, department name, manager, and number of employees.

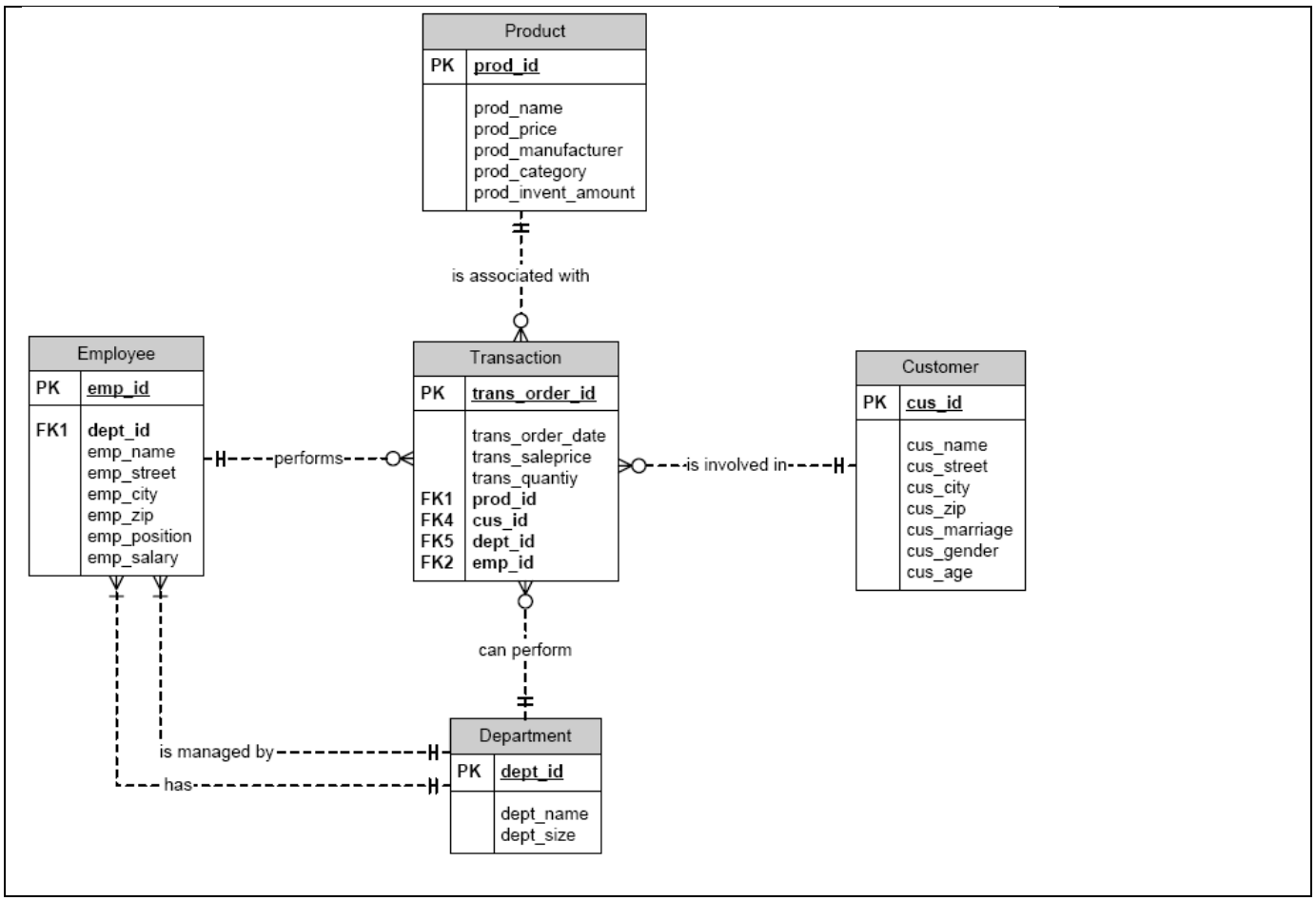
Each department should have at least one manager, who should also be an employee in that department. The company does not support joint appointments, which means that employee can only work in one department holding one position. Each transaction can be performed by one salesperson and it can be associated with only one customer. Each product has a regular price, as well as a sale price, which may be different during the purchase transaction.

4. Identify all possible business rules for the above case study. (6 marks)

***Key business rules*:**
1. Each department should have at least one manager; a manager can manage only one department.
2. Each department can have at least one employee. Each employee can work only in one department.
3. Each transaction can be performed by only one salesperson; each salesperson can perform no or many transactions.
4. Each transaction is associated with only one customer; each customer can be involved in no or many transactions.
5. Each transaction is associated with only one product; each product can be associated with nor or many transactions.
6. Each transaction corresponds to only one department; one department can perform no or many transactions.

5. Create an ERD showing a possible database design. Show the entities, relationship between the entities (show the connectivity, cardinality, and strength of relationship), primary and foreign keys, and other attributes. (16 marks)

   Note the following options are provided to create the ERD:
   A. An ERD created using MS Visio, if you have MS Visio installed on your PC/laptop, or
   B. An ERD created using any third party ERD diagramming tool, or
   C. An ERD created using MS Word's Shapes tools, or
   D. A hand drawn ERD which can be photographed and inserted in this document.



Normalization (30 marks)

The following data was previously maintained in a spreadsheet. You are required to normalize the data in the time sheet from 1NF to 2NF to 3NF by answering questions 6-8.

Note:

- To draw the dependency diagrams, you can use MS Word's table, shapes and drawing tools.

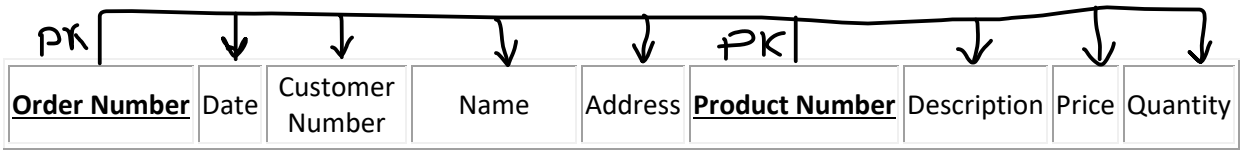
Order Number	Date	Customer Number	Name	Address	Product Number	Description	Price	Quantity
1	5/11/2004	22	Alan Smith	London	A95	Jacket	55	4
					G17	Coat	120	8
					K10	Suit	90	5
2	19/11/2004	47	Jay Jones	Paris	G17	Coat	120	9
					D77	Shirt	35	20
3	27/11/2004	25	Sally West	Glasgow	E30	Tie	5	25
					D77	Shirt	35	4

6. 1NF describes tabular format in which a) all key attributes are defined, b) there are no repeating groups in the table, and c) all attributes are dependent on the primary key.

   a) Convert the above data into a tabular format that satisfies 1NF. *(5 marks)*

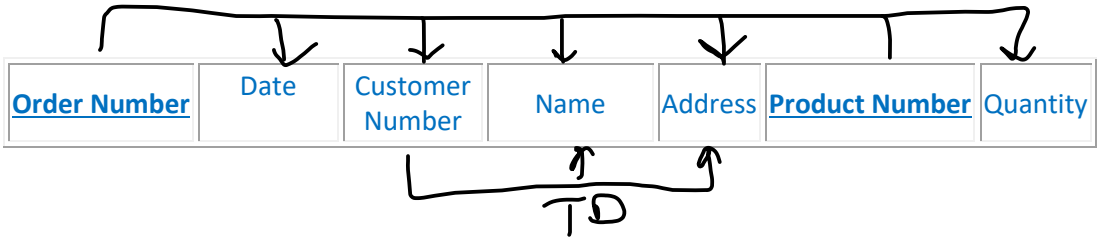
Order Number	Date	Customer Number	Name	Address	Product Number	Description	Price	Quantity
1	5/11/2004	22	Alan Smith	London	A95	Jacket	55	4
1	5/11/2004	22	Alan Smith	London	G17	Coat	120	8
1	5/11/2004	22	Alan Smith	London	K10	Suit	90	5
2	19/11/2004	47	Jay Jones	Paris	G17	Coat	120	9
2	19/11/2004	47	Jay Jones	Paris	D77	Shirt	35	20
3	27/11/2004	25	Sally West	Glasgow	E30	Tie	5	25
3	27/11/2004	25	Sally West	Glasgow	D77	Shirt	35	4

   b) Create a dependency diagram clearly showing partial and/or transitive dependencies (if any). The identified primary key(s) (PK) must be clearly shown. Any assumptions made must be stated clearly. *(5 marks)*



7. A table is in 2NF when it, a) is in 1NF and b) includes no partial dependencies. Using your solution from question 6,

   a) showing the dependency diagrams for each table in 2NF. *(5 marks)*

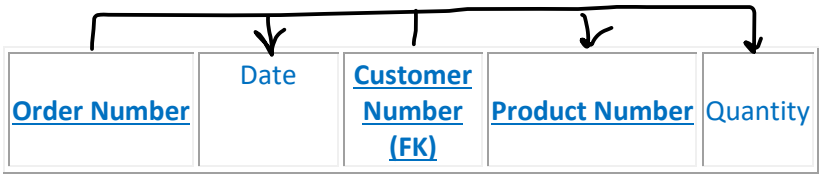


   b) write the relational schema for tables in 2NF. *(5 marks)*

   Order (**Order Number**, Date, Customer Number, Name, Address, **Product Number**, Quantity)
   Product (**Product Number**, Description, Price)

8. A table is in 3NF when it, a) is in 2NF and b) includes no transitive dependencies. Using your solution from question 7,

   a) showing the dependency diagrams for each table in 3NF. *(5 marks)*



b) write the relational schema for tables in 3NF. *(5 marks)*

Order (**Order Number**, Date, **Customer Number**, **Product Number**, Quantity)
Product (**Product Number**, Description, Price)
Customer (**Customer Number**, Name, Address)

Note: The below answer can be accepted as well if refined further.

Refining the 3NF for Order relation we get the following schemas:
Order (**Order Number**, Date, **Customer Number (FK)**)
OrderLine (**Order Number**, **Product Number**, Quantity)
Product (**Product Number**, Description, Price)
Customer (**Customer Number**, Name, Address)

End of Part B